

Search Ad Optimization — Research Update

February 19, 2026

Implementation Progress

Google Ads API Integration

Currently focusing on automating the inputs and outputs through the Google Ads API:

Inputs (downloading data):

- Keyword-level daily performance (clicks, costs).
- Campaign-level performance (clicks, purchases, by segment).
- Historical keyword metrics (search volume, competition).
- Previously downloaded manually; now being automated for daily retrieval.

Outputs (uploading solution):

- Campaign daily budgets (by region $\times$ match type).
- Base bids for each keyword-match type-region combination.
- Bid adjustments (hour of day, device, age, location).

LLM Keyword Scoring

Alternative to BERT embeddings

Motivation: Replace generic BERT embeddings (768-d $\rightarrow$ 50-d via SVD) with a domain-specific relevance score from an LLM.

Approach:

- Use Qwen3-8B to score each keyword on a 1–5 relevance scale.
- The score replaces the 50 embedding dimensions with a single feature.
- Scoring is based on a rubric that accounts for purchase intent, learning intent, and irrelevance signals.

Prompt structure (keyword = “machine learning course”, course = ML):

- 1 Start with Base Score = 3.
- 2 -1 if keyword contains “free”, “cheap”, “youtube”, “login”, or is unrelated.
- 3 -1 if purely informational (“what is”, definitions, news).
- 4 +1 if implies learning intent (“course”, “training”, “tutorial”).
- 5 +1 if implies high purchase intent (“certification”, “bootcamp”, “university”).

Apply all modifiers cumulatively. Score clamped to [1, 5].

BERT vs. LLM Scores

Gen AI course, Dec 1–31 backtest

Experiment	Optimized (daily avg)					Improvement (vs. Baseline)		
	Clicks	Purch	Cost	Clicks/\$	Kws	Clicks	Purch	Clicks/\$
BERT	242.5 ± 11.5	0.16 ± 0.01	362.83	0.668	101	+31.2%	+42.3%	+25.8%
LLM	176.8 ± 7.5	0.11 ± 0.00	362.84	0.487	53	+6.0%	+7.2%	+1.7%

Key takeaway: BERT embeddings substantially outperform LLM relevance scores in this backtest setting.

BERT vs. LLM Scores

Model quality comparison

Experiment	MSE	R^2
BERT	4.40 ± 0.04	0.927 ± 0.000
LLM	5.44 ± 0.11	0.913 ± 0.001

Averaged over 31 backtest days (Dec 1–31).

Discussion:

- BERT provides 50 embedding dimensions capturing semantic variation among keywords; the LLM approach compresses this to a single 1–5 score.
- The LLM score is more interpretable but the existing set of keywords all have similar scores.
- Could be useful in the future when we have results for a broader set of keywords with varying LLM scores.

Next Steps

- ➊ **Continue implementation:** Google Ads API integration for automated daily data download and bid upload.
- ➋ **Fine-tune BERT embedding dimensions:** Currently using 50 SVD components; experiment with different values (e.g., 20, 100, 200) to find the optimal dimensionality.
- ➌ **Add robustness via ensemble models:** Train multiple XGBoost models (with different bootstrap samples) and average their predictions, as described in the reference paper.
- ➍ **Validate bid multiplier:** The $\times 1.3$ second-price auction adjustment needs empirical validation once live bidding begins.